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Vacuum Assisted Closure Improves the Quality of Life in Patients with Diabetic Foot

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Abstract. *Background :* Diabetes Mellitus (DM) is the most common endocrine disease worldwide. One of the most important chronic complications of this disease is the development of diabetic foot. The management of diabetic foot wounds is quite important with respect to public health.
Aims : To determine the effect of Vacuum Assisted Closure (VAC) therapy on the quality of life in the treatment of diabetic foot ulcers and compare it with standart wound care.
Methods : Between May 2007 to December 2008, 67 consecutive patients with diabetic foot ulcers were randomly assigned to VAC therapy (Group1, n : 30) or standart wound care (Group 2, n : 37). The SF-36 questionnaire was administered the day before and in the month following wound healing. Global analyses of the 8 domains and 2 comprehensive indexes of SF-36, Physical Component Summary (PCS) and Mental Component Summary (MCS) were performed. Clinical measures included standard antidiabetic treatment, daily wound care including antiseptic bath, debridement, toe removal for gangrene when necessary, and wound care with conventional methods or VAC. Healing time was calculated as the time from hospital admission to the time of re-epithelization.
Results : There were no differences in the mean age, ulcer size and pulse status of the patients in both groups. Healing time in the VAC group was significantly reduced ($p < 0.05$). All 8 domains of SF-36 and MCS and PCS scores improved remarkably after VAC therapy.
Conclusion : Vacuum Assited Closure therapy was found to be effective in the treatment of chronic diabetic ulcers. The improvement of quality of life demonstrates a clear-cut indication in this particular group of patients.

Introduction

Diabetes Mellitus (DM) is a syndrome characterized with hyperglycemia resulting from absolute or relative impairment in insulin secretion and/or insulin action. In the course of the syndrome ; chronic hypergylcemia, disorders of carbohydrate, protein and fat metabolisms, capillary membrane changes and accelerated arteriosclerosis may occur. It is the most common endocrine disease worldwide. The prevalence of chronic complications has increased with the increased human life expectancy with the invention of insulin and other anti-diabetic agents. These complications are the main causes of mortality and morbidity among diabetic patients (1). Prevalence of DM was detected to be 7.2% in our country (new 2.3%, known 4.9%) (2). In the fact of the highest prevelance in the world and the heaviest complications occuring during this syndrome, DM is the first disease to recieve early diagnosis. About 100 million people suffered from DM in the world by the year 2000. It is expected to reach 220 million in the year 2010 and 300 million in 2025 (3, 4).

The diagnosis of DM may sometimes be delayed in our country as in some other even very well developed countries. Usually the patient lives unaware of the disease and presents with complications due to ignorance of the symptoms. The chronic complications are consequences of various effects of the DM. The correlation between blood glucose level and chronic complications has been shown in various studies. Nephropathy, neuropathy and retinopathy are the microvascular complications where as cerebrovascular diseases, ischemic heart disease and peripheric arterial diseases are the macrovascular complications of DM (5). The well known diabetic foot is one of the most important macrovascular complications of DM. Considerable long-term medical and economic consequences of all of these complications demonstrate the importance of DM.

The aim of the current study is to compare the effecacy of classical wound care and vacuum assisted closure

methods in patients with diabetic foot from the points of patient comfort and quality of life.

Materials and methods

The patients were diagnosed with DM according to the American Diabetes Association (ADA) diagnostic criteria. The patients with diabetic foot, who were admitted to our department between May 2007 and December 2008, were included to the study. Patients were informed about the study at the beginning and then they were requested to fill the Turkish version of quality of life questionnaire SF-36. Diabetic foot care technique was selected by random sampling.

Patients

Consecutive 67 patients admitted to our clinic with a diagnosis of diabetic foot between the dates May 2007 and December 2008, were randomized and divided into two groups as Group 1 (having VAC treatment n : 30) and Group 2 (conventional wound care treatment n : 37). Randomization of the patients was arranged by the free use web based system (http://www.tufts.edu/~gdallal/PLAN.HTM).

As a standard, all patients filled in the SF-36 questionnaire form at the beginning of the treatment. They filled the questionnaire once again in the month following wound healing, mean in the 4th month (2-8 months). SF-36 form shortly consisted of 8 data and 2 comparative sections. The first section of the questionnaire compared the physical state and the second section compared the mental state.

Physical State :

Physical Function (PF) : Whether or not restrictive in performing daily activities.
Physical role (PR) : Whether or not having problems with work or other daily activities as a result of physical health.
Bodily Pain (BP) : Assessment of pain.
General Health (GH) : General health assessment.

Mental State :

Vitality (VT) : How much energy does the patient feel?
Mental Health (MH) : How does the patient feel (depressed, happy, nervous...etc)?
Social Function (SF) : Whether or not interfer with normal social activities due to physical and emotional problems.
Role-Emotional (RE) : Interruption of working and daily activities due to emotional state.

Anti-diabetic treatment, daily wound care, debridement and the treatment of gangrenous tissue when required were standard for all patients. Group 1 patients were treated with VAC in addition to the standard treatment. A comparison of the mean recovery period, duration of epithelization and responses to SF-36 were analyzed.

Statistical analysis

All the data were assessed using SPSS (Statistical Package for the Social Sciences) 16.0 program pack (SPSS for Windows, SPSS Inc, Chicago, IL). Variables were expressed as mean ± standard deviation. Differences between continuous variables before and after treatment were compared with ANOVA and Kruskal-Wallis test. The data of SF-36 were assessed using non-parametric tests. Mann-Whitney U test was used to compare SF-36 form between conventional wound care and VAC. SF-36 physical and mental components were compared separately for two groups within each group. Differences were considered statistically significant for a p value of < 0.05.

Results

Demographic data (age, gender, type of diabetes, wound diameter) of patients were presented on Table I. There were 10 males and 20 females in Group 1 where as 9 males and 28 females in Group 2. Age ranged between 57-79 (68.5 ± 11.1) years in the VAC group and 54-78 (66.3 ± 12.6) years in the conventional treatment group. Most of the patients had Type I DM (> %99) in both groups and received insulin therapy ; hoever, despite diet and medications, DM control was majorly poor in patients in both groupd determined with HbA1C levels. Duration of DM in average was 11.3 ± 9.2 years in Group 1 and 9.3 ± 7.6 years in Group 2 ; and foot ulcers were present for 11.3 ± 9.2 weeks and 8.8 ± 7.2 weeks in Group 1 and Group 2, respectively. The ulcers sized 35.7 ± 6.4 cm^2 in the VAC group and 29.7 ± 5.2 cm^2 in the conventional treatment group. Ulcer bed was infected in 24 patients, edematous in 5 patients, and relatively clean in 1 patient in Group 1. In Group 2, ulcer bed was infected in 30 patients, edematous in 5 patients, and relatively clean in 2 patients. On the other hand the ankle-brachial indices were above 0.7 in 28 patients in the VAC therapy group and 34 patients in the conventional treatment group. Twenty-eight patients had depressive mood state in Group 1 and the respective was 32 patients in Group 2 before the treatment and the mood change to treatment was dramatic in both groups. The age, gender, duration of diabetes, and wound diameter were compared and the differences were found statistically insignificant (all the $p > 0.05$, detailed p values are presented on Table I). Both randomized groups were found to have homogenous distribution.

Table I

Demographic data of the patients

Characteristic	Standard Woud care (n = 37)	VAC (n = 30)	P value
Gender (m/f)	9 / 28	10 / 20	0.45
Age	66.3 ± 12.6 (54-78)	68.5 ± 11.1 (57-79)	0.56
Duration of DM	9.3 ± 7.6 years	11.3 ± 9.2 years	0.45
Ulcer duration	8.8 ± 7.2 weeks	11.3 ± 9.2 weeks	0.67
Ulcer size	29.7 ± 5.2 cm^2	35.7 ± 6.4 cm^2	0.65
DM Type (I/II)	35/2	29/1	0.75
Current antidiabetics Oral Insulin Diet	 1 35 1	 30 	0.65
Diabetes Control (HbA1C check) Good (steady) Fair (occasionally fluctuating) Poor (fluctuating)	 1 7 29	 2 28	0.06
ABI ABI > 0.7 ABI < 0.7	 34 3	 28 2	0.07
Ulcer bed Infected with slough Edematous with patchy necrosis Relatively clean	 30 5 2	 24 5 1	0.65
Gangrenous tissue Dry Wet None	 3 5 29	 1 2 27	0.274
Nutritional state Body weight	 62.1 ± 14.4 kg	 62.8 ± 14.5 kg	0.06
Depressive Mood Before Treatment After Treatment	 32 4	 28 2	 0.08 0.39

(m, male ; f, female ; DM, diabetes mellitus ; ABI, ankle-brachial index).

The 8 parameters were assessed with SF-36 health questionnaire. Physical and mental health states were checked with different questions. These parameters were analyzed separately among themselves. According to the SF-36 health questionnaire The VAC treatment indicated a significantly positive effect on both mental ($p = 0.0287$) and physical ($p = 0.004$) health when compared with the conventional wound dressing.

The changes in mean wound diameters of patients were also compared. The VAC treatment was determined to heal the wounds much better than conventional wound dressing during the same period of time (Table II).

Discussion

In the current study, patients with diabetic foot were examined with respect to adaptation on different wound care techniques. The efficiency of these wound care techniques over wound healing and sociologic status of the patients were observed.

The increase in the number of elderly people in the world population is accompanied with numerous problems. More than 30 million people constituted the population aged over 65 in the USA in 1995 and it is expected to exceed 60 million by 2030. Approximately 5000 people join the population of people over 65 years of age everyday. The average age will be approximately 85 years in 2030 (6). Improvement of the survival rates increases the mean age of diabetic patients and the risk of developing diabetic neuropathy and diabetic foot ulcer development is directley related to diabetic neuropathy (7). Diabetic foot ulcer is observed three times more than ischemic ulcers. Approximately 2 million of 14 million patients with DM in the United States will

Table II

Duration of granulation

Group	Time to healing (weeks)
	Group I
Mean SD	5.3 ± 1.4
Median	4.4
	Group II
Mean SD	4.2 ± 1.9
Median	3.9
P value	< 0.05

develop diabetic foot ulcer. Amputation is performed on diabetic patients 12 times more commonly than the non-diabetics. About 5-15% of diabetic patients may undergo amputation. More than 54,000 diabetic patients were observed to undergo non-traumatic amputation in the United States. The length of hospital stay for patients with diabetic foot wound is longer than that of for normal population. The mean survival rate was found to be 50% in these patients (8, 9). Small sized wounds are observed 15 times more in diabetic patients and diabetic foot was determined to be 50% preventable with a regular treatment, debridement and limited amputations when the ulcers start to occur (9, 10). Considering the duration of hospital stay, medical expenses, morbidity and mortality, the treatment of diabetic foot wound is very important with respect to public health.

The treatment of diabetic foot wounds has various steps and should be handled with multi disciplinary approach. These are the wound assessment (etiology, size, depth ...etc), infection treatment, collection of bacteria cultures, radiological studies (Charcot's foot, foreign object, subcutaneous air, ...etc), metabolic control, detection of vascular state, close follow-up and wound care following studies including all above (11). Diabetic foot wound care is not fundamentally different than the other methods of wound management. There are numerous studies performed on diabetic wounds in the literature. The effects of various wound care tecniques on wound healing were compared in these studies ; however, there is not a current consensus on an optimal wound care technique in this particular patient group (11, 12). Generally, conventional wound care is performed. Conventional wound care should be performed as follows : wound cleansing and debridement, elimination of necrotic tissue and wound closure using sterilized gauze dressing. This is a quite practical, effective, cheap and reliable method. The major disadvantages are inability to perform wound dressings on time due to the adaptation problems of diabetic patients and the requirement for the repetition of this conventional treatment during the day in exudative wounds. Repetitive wound care is performed regularly at hospital conditions ; however, this procedure is not applicable for outpatients. Wound care techniques have improved parallel to the technology and various wound care products have been launched on the market. These wound care products are sold under various groups such as hydrocolloids, hydrogels, films, foams and Ca alginate. On the other hand, there is no wound-care product proven to spur and shorten duration of wound healing, reduce cost and the length of hospital stay up to date. The most important phase of wound care is elimination of the necrotic tissue from the wound, which basically contributes to wound healing. Debridement surgery should be performed at the beginning of wound treatment. It should be preferred especially in the treatment of exudative and necrotic diabetic foot wounds. Performance of additional debridements to such wounds during conventional wound care period reduces the duration of wound healing and the length of hospital stay and also increases antibiotic penetration. Wound debridement is generally performed as surgical wound debridement in the clinics although chemical agents, wound care products and equipments executing wound debridement have been lately produced. Among them the most important is the vacuum-assisted apparatus for the debridement of wounds. This apparatus is revolutionary in wound care especially in exudative wounds requiring debridement (13, 14, 15, 16, 17).

In our study, although the conventional wound care of patients with a wound infection and exudation was performed twice daily on average, epithelization of the wound took longer than patients who had VAC treatment for wound care. In parallel, hospital stay was longer in Group 2 than Group 1. When assessing the effects of wound dressing techniques on quality of life, the Turkish version of quality of life SF-36 form was used. In this form there were questions interrogating the patients states of physical and mental health. The statistical assessment showed that physical and mental health parameters were significantly improved in patients with VAC treatment compared to the conventional wound dressing treatment. The superiority of VAC treatment over conventional wound dressing treatment may be summarized as : performance of once-daily wound dressing, decrease in the frequency of patient disturbance for the management of debridement, feeling less anxiety because of exudation and comfortable movements of patient. Lack of movement restriction especially in physical parameters is remarkable as the major positive characteristic of VAC treatment.

In conlusion, diabetic foot wound is an important health problem in our country. Patients frequently confront both mental and physical health problems. The VAC treatment provides positive effects over wound healing as well as quality of life in wound management by promoting mental status of the patients and inducing

physical movement ability. It is safe, reliable and applicable in suitable patients.

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